

Chapter 3 — In the Earliest Cities

The Harappan / Indus Valley Civilisation

The one-line point: ~4,700 years ago, people on the Indus built **planned cities with grid streets, covered drains, standardised bricks and public baths** — and then, ~800 years later, it all ended and nobody knows exactly why. **This is the highest-yield chapter in the book.**

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Syllabus Classification

Field	Value
Subject	History → Ancient India
Sub-discipline	Proto-history / Bronze Age Urbanism (proto-history = writing exists but is undeciphered)
Period	~2600 BCE – 1900 BCE (mature phase); ~4,700 to ~3,900 years ago
Prelims (GS Paper I)	<i>History of India</i> → the single most-asked ancient India topic
Mains (GS Paper I)	<i>Indian culture — art forms & architecture from ancient times</i> : town planning, seals, sculpture, drainage
Also feeds	Art & Culture (seals, Dancing Girl, Priest-King, Pashupati seal), Geography (river systems, Ghaggar-Hakra), Environment (water management, urban decline & climate), Governance (sanitation, urban planning as a model)
Weightage	★★★★★ — asked in roughly every alternate Prelims ; multiple Mains appearances
Overlaps with	Current affairs : Rakhigarhi DNA studies, Dholavira's UNESCO World Heritage inscription (2021), Aryan migration debate

 **Not Modern History. This is Ancient India → Proto-history**, and it crosses heavily into **Art & Culture**. If you learn one chapter from this book properly, make it this one.

Learning Outcomes

By the end of this chapter you should be able to:

- ☐ **Describe** the standard Harappan city layout — citadel vs lower town — and say what each contained.
- ☐ **Explain** the drainage system in technical detail and argue why it implies a central authority.
- ☐ **Locate and match** all major sites to their unique features: Mohenjodaro, Harappa, Lothal, Dholavira, Kalibangan, Chanhudaro, Rakhigarhi, Banawali, Surkotada, Nageshwar, Balakot, Shortughai.
- ☐ **List** the crops, animals and technologies, and identify what the Harappans **did not** have.
- ☐ **Explain** why the seals and script matter, and state precisely what we can and cannot conclude from an undeciphered script.
- ☐ **Trace** the raw-material trade network — which material came from where — and name the Mesopotamian trade terms (Meluhha, Magan, Dilmun).
- ☐ **Explain** the weight system and why standardisation is evidence of centralised control.

- ☐ **Evaluate** the competing theories for the civilisation's decline, without overcommitting to one.
- ☐ **Compare** Harappan and Vedic cultures point-by-point (this is a recurring Prelims format — see Ch.4).
- ☐ **Deploy** Harappan town planning in a Mains answer on ancient Indian urbanism or water management.

Start Here: What "First City" Actually Means

You've read about villages in Chapter 2 — a few hundred people, mud huts, everyone farming.

Now picture this instead:

A city of 35,000–40,000 people. Streets laid out on a grid, crossing at right angles. Every house connected to a covered drain running beneath the street, with inspection holes for cleaning. Bricks in a fixed 1:2:4 ratio, identical from Gujarat to Afghanistan — a distance of over 1,500 km. A public water tank made watertight with gypsum mortar and bitumen. Standardised stone weights in a binary sequence.

Now the part that should genuinely unsettle you: **this was 4,700 years ago.** Many Indian towns did not have covered sewerage in 1947.

WHO ELSE WAS BUILDING CITIES AT THIS MOMENT? (~2600 BCE)	
MESOPOTAMIA	– Sumer, Ur, Uruk (Tigris–Euphrates)
EGYPT	– Old Kingdom, the Great Pyramids
HARAPPA	– Indus valley
CHINA	– a bit later (Shang, ~1600 BCE)
THE HARAPPAN WAS THE LARGEST OF ALL OF THEM BY AREA – roughly 1.5 million sq km, ~1,500 km east-to-west	

And here's the twist that defines the whole chapter: Egypt gave us pyramids and pharaohs' names.

Mesopotamia gave us kings and epics. **Harappa gave us drains, weights and grids — and not one single ruler's name.** We cannot read a word they wrote.

The Software Engineer's Analogy

The Harappan civilisation is **a large-scale distributed system whose source code we possess but cannot decompile.**

WHAT WE HAVE	WHAT IT TELLS US
Identical brick ratio 1:2:4 across 1,500 km	→ A SHARED SPEC, centrally defined and enforced. Nobody achieves this by accident.
Standardised weights in 1,2,4,8,16,32... sequence	→ A STANDARDS BODY. Trade needs agreed units. Someone published and policed the standard.
Grid streets + covered drains laid BEFORE houses	→ CENTRAL PLANNING, and planning BEFORE building. Infrastructure first, then the app layer.
Seals with a script we cannot read	→ An ENCODED PROTOCOL we have packets for but no schema. ~4,000 samples, avg 5 signs each – too little data to crack.
NO palaces, NO grand tombs, NO royal names	→ Either a very different power structure, or one that left no monuments to itself. UNRESOLVED.

🔑 **The takeaway to carry into every Harappan question:** we have extraordinary evidence about their *engineering* and almost none about their *politics or religion*. **So be very careful about claims regarding Harappan rulers, gods or warfare** — that's exactly where Prelims sets its traps.

1 Discovery — How We Found Them

This is a genuinely good story, and the names are asked.

TIMELINE OF DISCOVERY

- 1850s → British engineers building the Lahore-Multan RAILWAY find a mound of high-quality ancient bricks at Harappa. They GRIND THEM UP and use them as track ballast. An unknown quantity of the city is gone forever.
- 1875 → Alexander Cunningham (first Director-General of the ASI) reports a seal from Harappa. He cannot place it, and wrongly assumes it is foreign.
- 1921 → DAYA RAM SAHNI excavates HARAPPA (Punjab, Pakistan)
- 1922 → R. D. BANERJI excavates MOHENJODARO (Sindh, Pakistan)
- 1924 → JOHN MARSHALL (ASI Director-General) announces to the world that a previously unknown civilisation has been found. India's history is instantly pushed back ~2,000 years.
- Post-1947 → Partition puts Harappa and Mohenjodaro in Pakistan. Indian archaeologists search for sites in India and find LOTHAL, KALIBANGAN, DHOLAVIRA, RAKHIGARHI, BANAWALI...
- 2014-2025 → RAKHIGARHI (Haryana) emerges as possibly the LARGEST Harappan site; ancient DNA studies from here become central to the Aryan migration debate.
- 2021 → DHOLAVIRA inscribed as a UNESCO WORLD HERITAGE SITE

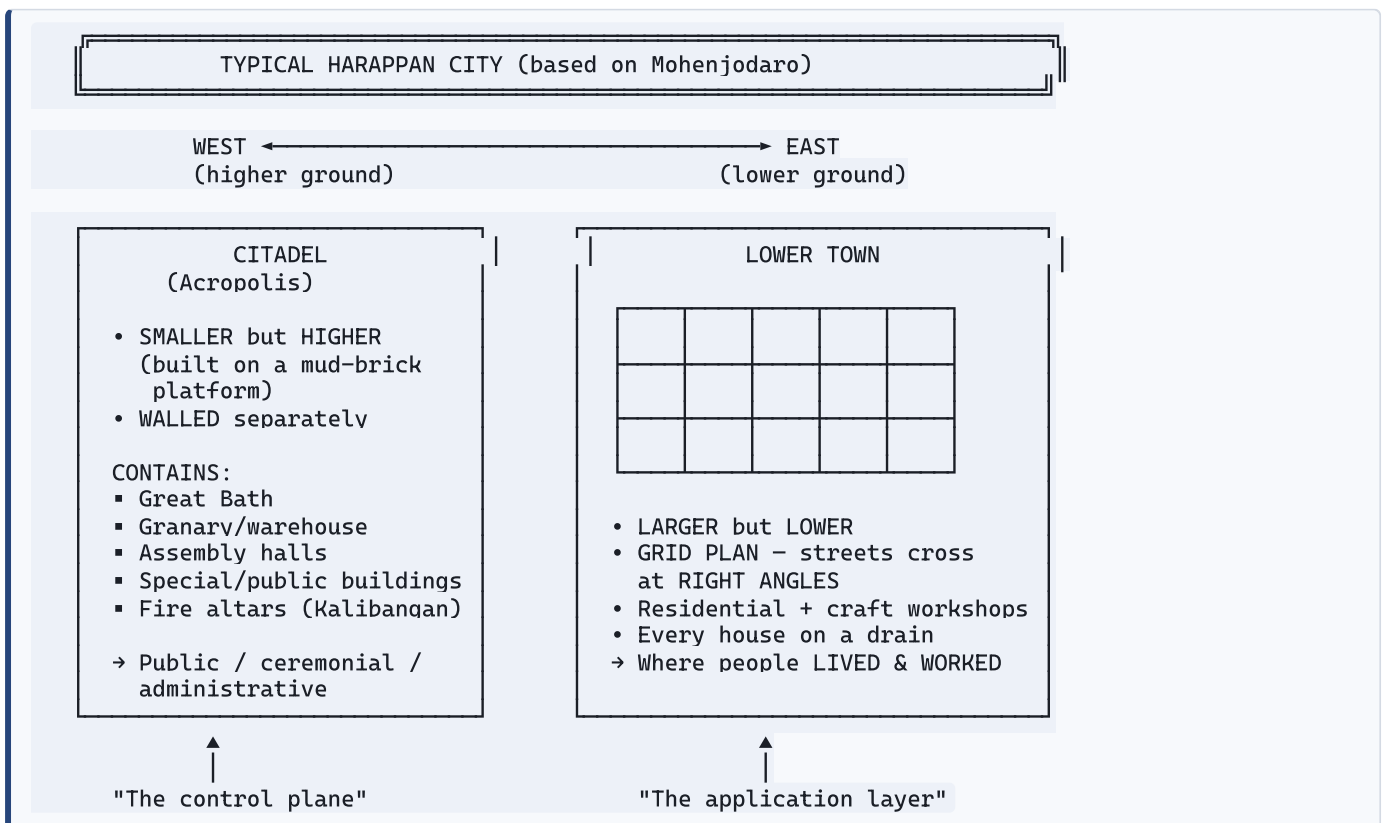
The naming question — "Harappan" or "Indus Valley"?

Name	Why used	Why it's imperfect
Harappan Civilisation	Archaeological convention: name a culture after the first site excavated — Harappa (1921)	Harappa wasn't the capital or the biggest site
Indus Valley Civilisation	Most early sites lie along the Indus and its tributaries	Misleading — many sites (Lothal, Dholavira, Kalibangan, Rakhigarhi) are <i>not</i> in the Indus valley. Over half lie along the dried-up Ghaggar-Hakra river
Indus-Saraswati Civilisation	Used by those identifying Ghaggar-Hakra with the Rigvedic Saraswati	Contested; the identification is debated

🎯 **Prelims-worthy nuance:** "Harappan Civilisation" is the **more accurate** term precisely because the civilisation extends well beyond the Indus valley. NCERT uses "Harappan."

2 The City Layout — The Master Diagram

Draw this from memory. It underpins half the questions on this chapter.



The rules of Harappan urban planning

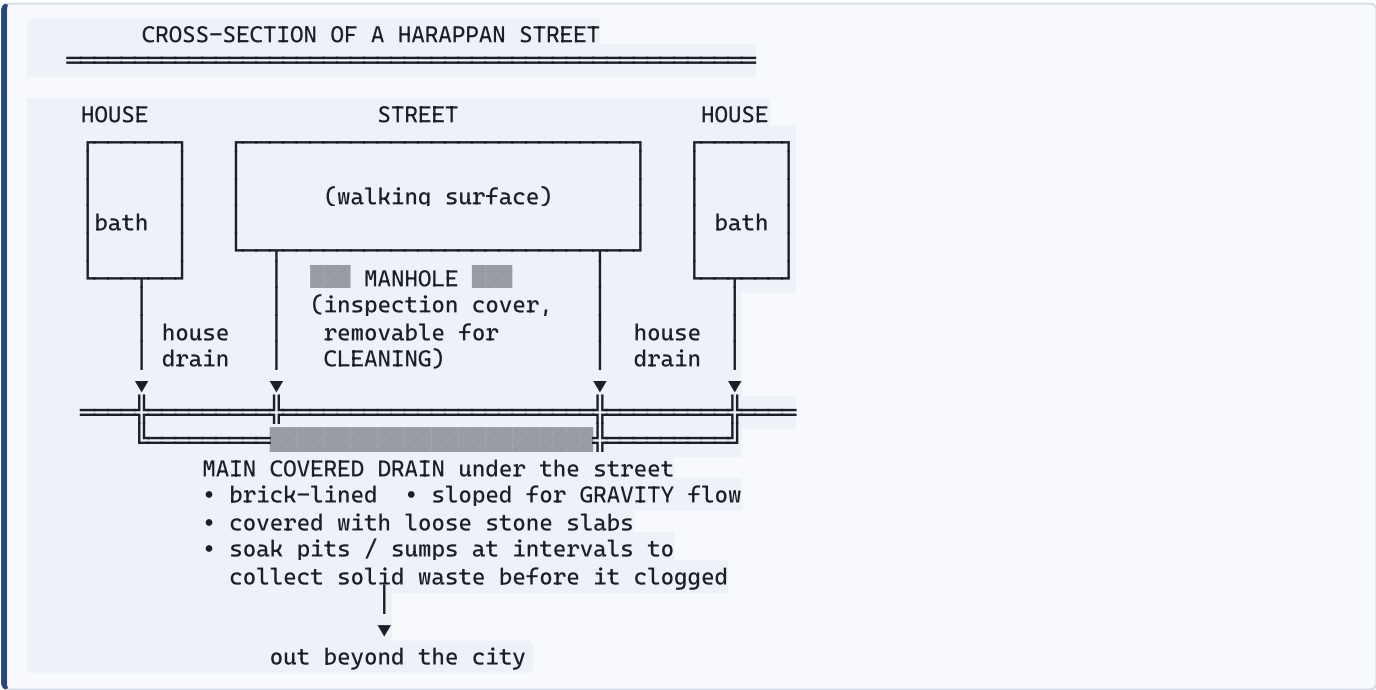
Feature	Detail
Two-part division	Citadel (west, smaller, higher, walled) + Lower Town (east, larger, lower, walled). Both usually walled
Grid street plan	Streets and lanes cross at right angles , dividing the city into rectangular blocks
Baked bricks	Standardised and remarkably well-made; laid in an interlocking pattern for strength
Brick ratio	1 : 2 : 4 (thickness : width : length) — the <i>same ratio</i> across the entire civilisation
Drains first	Drains were laid out BEFORE the houses were built — the killer detail
Wells	Many houses had their own well ; Mohenjodaro had ~700 wells
No temples or palaces	No structure has been <i>conclusively</i> identified as either

★ **"Drains were planned before houses" is the single most quotable line in this chapter.** Think about what it requires: someone surveyed the site, designed a network, dug and built it, and only then allowed construction — and everyone complied. **That is a state, or something very like one.**

🖥️ You'd call it *infrastructure-as-code, provisioned before deployment*. You cannot retrofit a gradient-correct sewer network under an existing city. It must be designed first. So must a civilisation.

3 The Drainage System — Their Greatest Achievement

Take this seriously. It is what makes the Harappans exceptional.



Why this is genuinely remarkable

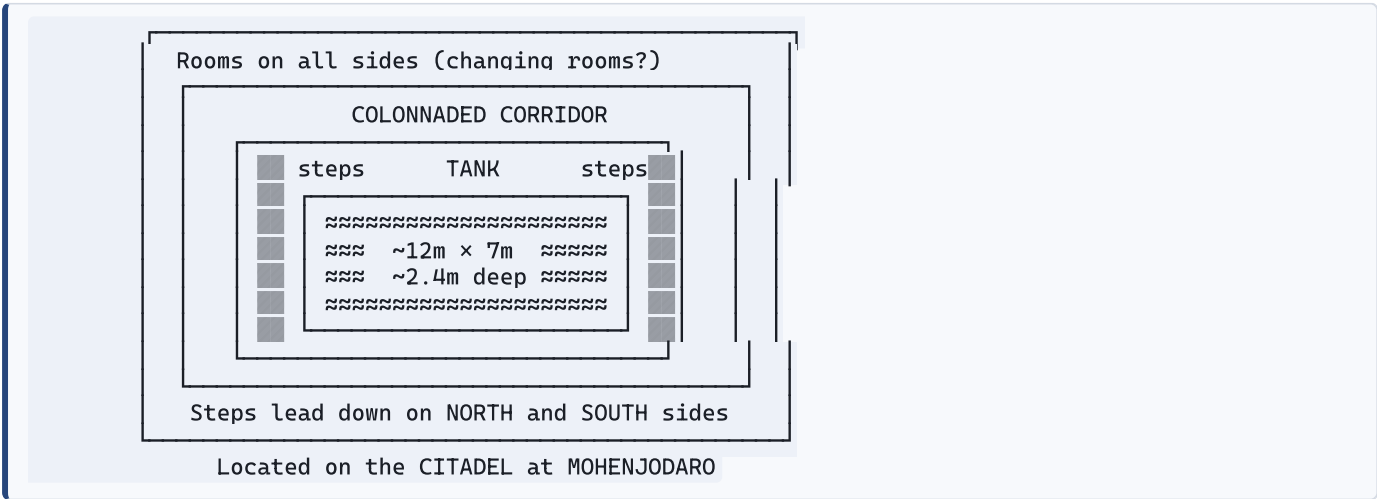
Feature	What it required
Every house connected	Universal coverage — not a privilege of the rich. Suggests a <i>civic</i> system, not an elite one
Covered drains	Understanding that open sewage causes disease and smell
Gradient/slope	Careful surveying — water must flow by gravity across the whole city
Manholes / inspection holes	Designed for maintenance , not just construction. Someone was responsible for ongoing cleaning
Soak pits / sumps	Solid waste settled out before it could block the line
Standard brick sizes	Manufacturing standards, and a supply chain to enforce them

 **They designed for observability and maintenance.** Manholes are access ports; soak pits are a filtering stage that prevents downstream failure. This isn't just building a thing — it's building a thing that can be *operated* for generations. Most ancient engineering doesn't do this.

The organisational implication (say this in a Mains answer): a system like this cannot be built or maintained by individual households acting alone. It requires **surveying, standardisation, coordinated construction, and continuous municipal maintenance** — in other words, a functioning **civic authority** with the power to plan and enforce.

4 The Star Structures

 **The Great Bath (Mohenjodaro)**
The most famous Harappan structure.



Detail	Value
Location	Citadel, Mohenjodaro
Size	~12 m long x 7 m wide x 2.4 m deep
Waterproofing	Bricks set on edge with gypsum mortar ; a layer of bitumen (natural tar) lining the sides and floor
Water supply	From a well in an adjacent room
Drainage	A large corbelled drain in one corner emptied the tank
Surroundings	Corridor on all four sides; rooms on three sides — one with a well
Purpose	Probably ritual bathing on special occasions — a public/ceremonial function

 **The Great Bath is your evidence for "ritual purity mattered."** But be careful: we're *inferring* ritual from architecture. It's the leading hypothesis, not a proven fact — and Prelims respects that distinction.

 **Why it's astonishing engineering:** they solved waterproofing 4,700 years ago using gypsum mortar plus bitumen — a two-layer system. That's a genuine materials-science achievement.

The Granary / Warehouse

Found at **Mohenjodaro** and **Harappa** (Harappa's is near the river, with circular brick platforms nearby, possibly for threshing grain).

- Large structures on the **citadel**, with **ventilation ducts** and raised platforms (keeps grain dry, deters rodents)
-  **Terminology caution:** these were identified as "granaries" by early excavators, but the identification is **disputed** — no significant grain residue was found in Mohenjodaro's. Many archaeologists now call them "warehouses" or "great halls." NCERT still uses "granary."

 **Why a granary on the citadel matters:** collecting and storing grain centrally implies **taxation or tribute** — someone gathered surplus from farmers and stored it under guard. This is the direct continuation of Chapter 2's "storage → surplus → the state" chain.

Fire Altars

Found at **Kalibangan** (Rajasthan) and **Lothal** (Gujarat).

- Rows of raised brick platforms with ash, charcoal and terracotta cakes
- At Kalibangan, found both on the citadel *and* inside private houses
- Interpreted as evidence of **fire rituals / sacrifice**

 **Why this is significant for a bigger debate:** fire ritual is central to *Vedic* religion (Chapter 4). Fire altars at Harappan sites are used by some scholars to argue for continuity between Harappan and Vedic culture. **Contested, but very much asked.**

The Dockyard (Lothal) — ★★ ★

Lothal, Gujarat — one of the most important sites for India.

Feature	Detail
The dockyard	A large brick-lined basin , ~214 m × 36 m, connected to the Bhogavo river/Gulf of Khambhat by a channel
What it did	Ships and boats came in; a sluice gate and spill channel maintained water level at low tide
Why it matters	The world's earliest known dockyard (as identified). Lothal was a port and trading hub
Also at Lothal	A bead-making factory (with kiln, raw material, finished and unfinished beads), a warehouse , fire altars , a Persian Gulf seal, and rice cultivation
Layout note	Unlike most sites, Lothal's citadel and lower town were enclosed within one wall

 **Lothal = trade + beads + dockyard + rice + fire altar.** Four or five separate "famous for" tags on one site. Very high frequency in Prelims.

Dholavira (Gujarat, Rann of Kutch) — ★★

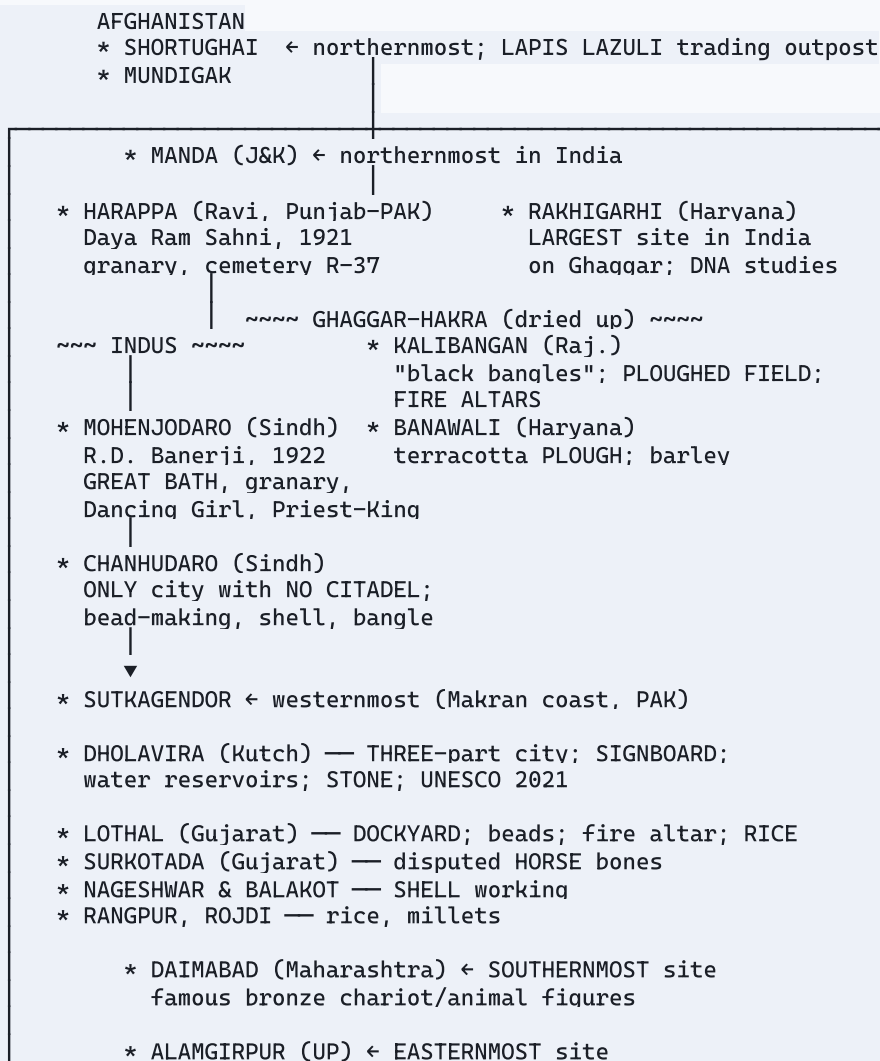
UNESCO World Heritage Site (2021) — which makes it current-affairs-relevant too.

Feature	Detail
Unique layout	Divided into THREE parts (not two): Citadel, Middle Town, Lower Town — each with massive stone walls
The signboard ★	A large inscription of ~ 10 large Harappan signs , made of white gypsum inlay on wood, found near the citadel gate — the world's earliest known "signboard"
Water management ★	An extraordinary system of stone-cut reservoirs , check dams and channels to harvest every drop of monsoon runoff — Dholavira sits in an arid zone between two seasonal streams (Mansar and Manhar)
Building material	STONE masonry (most Harappan sites are brick) — local stone was abundant
A "stadium"	A large open ground with terraced seating — possibly for ceremonies or public gatherings

 **Modern relevance:** Dholavira is regularly cited in water-conservation and rainwater-harvesting discussions. **Use it in Mains/Essay answers on water security** — an ancient Indian city in a desert that survived on engineered rainwater harvesting. That's a strong, specific example.

5 The Site Map — Learn Every One of These

HARAPPAN CIVILISATION – EXTENT & KEY SITES



📖 The extremities — a classic Prelims question format

Direction	Site	Location
Northernmost	Manda	Jammu & Kashmir (on the Chenab)
Northernmost overall (outpost)	Shortughai	Afghanistan (lapis lazuli)
Southernmost	Daimabad	Maharashtra (on the Pravara)
Easternmost	Alamgirpur	Uttar Pradesh (on the Hindon)
Westernmost	Sutkagendor	Makran coast, Baluchistan (Pakistan)

🔴 **Memory hook: "MDAS"** going N-S-E-W → **M**anda (N), **D**aimabad (S), **A**lamgirpur (E), **S**utkagendor (W).

The complete site table — your revision anchor

Site	Location	River	Excavated by	★ Famous for
Harappa	Punjab, Pakistan	Ravi	Daya Ram Sahni, 1921	First excavated; granary; cemetery R-37 ; workmen's quarters; bricks looted for railway
Mohenjodaro ★★★	Sindh, Pakistan	Indus	R. D. Banerji, 1922	GREAT BATH ; granary; Dancing Girl (bronze); Priest-King (steatite); Pashupati seal ; ~700 wells. Name = " <i>Mound of the Dead</i> " in Sindhi
Lothal ★★★	Gujarat	Bhogavo	S. R. Rao	DOCKYARD ; bead factory; warehouse; fire altars ; rice ; Persian Gulf seal
Dholavira ★★★	Kutch, Gujarat	Between Mansar & Manhar	J. P. Joshi / R. S. Bisht	THREE-part city ; SIGNBOARD ; water reservoirs; stone architecture; "stadium"; UNESCO 2021
Kalibangan ★★★	Rajasthan	Ghaggar	B. B. Lal / B. K. Thapar	"Black bangles" ; PLOUGHED FIELD (earliest known); FIRE ALTARS ; wooden furrow evidence
Chanhudaro ★★	Sindh, Pakistan	Indus	N. G. Majumdar	The ONLY major city WITHOUT a citadel ; bead-making, shell and bangle workshops; a "craft town"
Rakhigarhi ★★	Haryana	Ghaggar	Amarendra Nath	Largest Harappan site in India ; granary, cemetery; ancient DNA studies central to the Aryan debate
Banawali ★	Haryana	Ghaggar	R. S. Bisht	Terracotta model of a PLOUGH ; good quality barley; both citadel and lower town but <i>no grid</i> streets
Surkotada ★	Gujarat	—	J. P. Joshi	HORSE bones (heavily disputed); stone-covered burial
Nageshwar & Balakot ★	Gujarat / Baluchistan coast	—	—	SHELL working — bangles, ladles, inlay
Shortughai ★	Afghanistan	Oxus/Amu Darya	—	Trading outpost for LAPIS LAZULI
Sutkagendor	Makran coast, Pak	—	—	Westernmost ; port on the Iran route
Alamgirpur	Uttar Pradesh	Hindon	—	Easternmost
Daimabad	Maharashtra	Pravara	—	Southernmost ; famous bronze chariot and animal figurines
Manda	Jammu	Chenab	—	Northernmost in India
Ropar / Rupnagar	Punjab, India	Sutlej	Y. D. Sharma	First site excavated in independent India (1953); dog buried with human

🧠 Memory hooks:

- **Kalibangan** → *kali* = **black** → "black bangles"; and it has the **ploughed field** ("Kalibangan — the farmer **banged** the plough into the field")
- **Chanhudaro** → "**Chanhu** has **no** citadel" (the odd one out)
- **Lothal** → "**Lot** of trade" → dockyard
- **Dholavira** → three parts, and its **signboard** — "Dhola-**VEER**-a" the *bold* one with the big sign
- **Surkotada** → "**Surkotada** = **horse**" (disputed)

6 Life, Craft and Technology

Agriculture

Crops grown	Notes
Wheat, barley	The staples
Pulses (lentil, chickpea)	Widespread
Sesame, mustard	Oil seeds
Rice	At Lothal and Rangpur (Gujarat) — relatively rare
Cotton ★	The Harappans were among the world's first cotton growers . Greeks later called it " <i>Sindon</i> " — from Sindh
Millets	In Gujarat sites

Farming technology:

- **The plough** — evidence from two sites:
 - **Kalibangan**: an actual **ploughed field** with two sets of furrows **crossing at right angles**, showing two crops grown together ★
 - **Banawali**: a **terracotta model of a plough** ★
- Ploughs were of **wood**, so the ploughs themselves haven't survived — only the field marks and the model
- **Irrigation** was needed (the region is semi-arid): canals found at **Shortughai**, and water reservoirs at **Dholavira**

Animals reared: cattle (very important), sheep, goat, buffalo, pig. Also **dogs** and **cats** (footprints of both, and a famous brick with a dog chasing a cat).

 **THE HORSE QUESTION — know this precisely.** Horse remains at Harappan sites are **rare and disputed** (Surkotada is the main claim, and specialists disagree on whether the bones are horse or wild ass/onager). There is **no depiction of a horse on any Harappan seal**. By contrast, the horse is *central* to Rigvedic society (Ch.4). **UPSC has asked this (2013)**. The safe formulation: "*Harappans did not use horse-drawn chariots in warfare; evidence for the horse is scanty and contested.*"

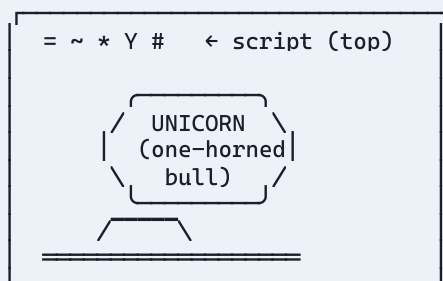
Crafts and materials

Craft	Detail
Bead-making ★	Carnelian (a beautiful red stone), agate, jasper, crystal, quartz, steatite, faience, shell, terracotta, gold. Chanhudaro and Lothal were bead factories
Faience ★	An artificial (man-made) material — ground sand/quartz mixed with colour and gum, shaped and fired to a glaze. Used for beads, bangles, small vessels. The Harappans manufactured a material that does not occur in nature
Shell working	Nageshwar and Balakot — bangles, ladles, inlay
Metallurgy	Copper and bronze (copper + tin). Also gold and silver . ❌ NO IRON — iron had not yet been discovered
Pottery	Fine wheel-made red pottery with black designs — plants, animals, geometric patterns
Textiles	Cotton and wool ; actual fragments of cotton cloth found; spindle whorls common
Seal-making	Steatite (soapstone), carved and then fired to harden
Stone tools	Chert blades, from the Rohri hills in Sindh

⚠️ **NO IRON.** This is a Bronze/Chalcolithic civilisation. Any statement claiming Harappan iron tools is **false**. Iron enters India only around 1000 BCE (Chapter 4/5). Easy Prelims elimination.

◆ Seals and the Script

The **seal** is the iconic Harappan artefact. ~**4,000 inscribed objects** have been found.



Typical seal:

- STEATITE (soapstone)
- Usually SQUARE, ~2–4 cm
- Animal motif in the centre
- Script above (avg 5 signs)
- Boss/knob on the back with a hole → worn or tied on
- Pressed into wet clay to leave a SEALING

Animals shown: the **one-horned "unicorn"** (most common), **humped bull**, elephant, rhinoceros, tiger, buffalo, goat, and composite/mythical creatures. **Notably absent:** the **horse**, and the **lion**.

Aspect of the script	What we know
Number of signs	Roughly 375–400 distinct signs
Type	Too many signs for an alphabet, too few for a purely logographic system → probably logo-syllabic
Direction	Generally right to left . Longer texts are sometimes boustrophedon — right-to-left on one line, left-to-right on the next (like an ox ploughing a field)
Length	Very short — average ~5 signs , longest ~26
Status	⚠️ UNDECIPHERED

Why can't we read it?

1. **No bilingual text** — nothing like the Rosetta Stone, which cracked Egyptian hieroglyphs by pairing them with Greek
2. **Texts are far too short** — 5 signs gives cryptanalysis almost nothing to work with
3. **We don't know the underlying language** — was it Dravidian? Indo-Aryan? Something extinct? Each assumption yields a different (unverifiable) decipherment
4. Even *whether it encodes language at all* is disputed by a minority of scholars

💻 **Precisely a cryptanalysis problem with a fatal data shortage.** You have ~4,000 ciphertext samples averaging 5 tokens each, no plaintext, no crib, and an unknown plaintext language. Frequency analysis needs volume and a known language model — you have neither. **The bottleneck isn't cleverness; it's data.** That's why 100 years of very smart people have failed.

What seals were probably FOR: stamping bundles of trade goods. Press the seal into wet clay on a package or a rope-tie; the sealing hardens; if it arrives intact, the goods weren't tampered with, and the mark identifies the sender.

💻 **A seal is a tamper-evident signature.** It doesn't encrypt; it *authenticates* and proves integrity in transit. Harappan sealings found in **Mesopotamia** are therefore direct evidence of goods travelling that far.

🏺 Weights and Measures

- Made of **chert** — a hard, uniform stone
- **Cubical**, mostly unmarked
- The sequence is **binary in the lower denominations: 1, 2, 4, 8, 16, 32, 64...** then shifts to decimal multiples of 16 (160, 200, 320, 640)
- **16 was clearly a key unit** ★ (a fun continuity: India used 16 *annas* to the rupee until 1957)
- The same weights turn up **across the entire civilisation**, from Gujarat to Afghanistan

📖 **A binary weight sequence isn't an accident.** With weights 1, 2, 4, 8, 16 you can compose *any* integer from 1 to 31 — the minimum set of weights for maximum coverage. That is a genuinely optimal design. Someone thought hard about this.

And the historical significance: standardised weights across 1.5 million sq km means **regulated, trusted, long-distance trade** — and an authority able to define and enforce a standard.

🎨 Famous artefacts (Art & Culture — know these by name and material)

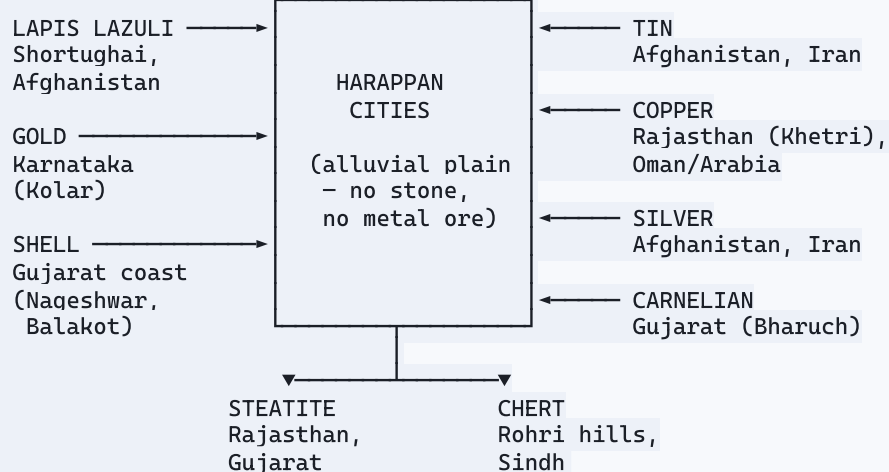
Artefact	Site	Material	Note
Dancing Girl ★★★	Mohenjodaro	Bronze (lost-wax / <i>cire perdue</i> casting)	~10.5 cm; a girl standing with hand on hip, bangles on one arm
Priest-King / Bearded Man ★★	Mohenjodaro	Steatite	Trefoil-patterned shawl over the left shoulder; half-closed eyes
Pashupati Seal ★★★	Mohenjodaro	Steatite	A seated yogic figure with a horned headdress, surrounded by elephant, tiger, rhino, buffalo, and deer beneath. Interpreted (contestably) as a proto-Shiva
Male Torso	Harappa	Red sandstone	Remarkably naturalistic
Bronze chariot & animals	Daimabad	Bronze	Famous hoard — chariot with rider, bull, elephant, rhinoceros
Mother Goddess figurines	Many sites	Terracotta	Numerous female figurines; often read as a fertility/mother-goddess cult
Terracotta toys	Many	Terracotta	Carts with movable wheels, whistles, animals, marbles — children existed ☺

🎯 **Learn the material, not just the name.** "Dancing Girl — bronze — Mohenjodaro" is a classic three-part matching question. So is "Priest-King — steatite" and "Male Torso — red sandstone."

7 Trade and Raw Materials — India's First Supply Chain

The core insight: the cities sat on flat alluvial plains that produce **grain but almost no stone, metal or gemstone**. Every hard material had to be imported. So the Harappans ran a continent-scale procurement network.

WHERE DID THE RAW MATERIALS COME FROM?



Material	Source
Copper	Rajasthan (Khetri mines); also Oman/Arabia
Tin ★	Afghanistan and Iran (rare — and essential to make bronze)
Gold	Karnataka (Kolar)
Silver	Afghanistan, Iran
Lapis lazuli ★	Afghanistan (which is exactly <i>why</i> Shortughai exists)
Carnelian	Gujarat (Bharuch region)
Steatite	Rajasthan, Gujarat
Shell	Gujarat coast — Nageshwar, Balakot
Chert (blades)	Rohri hills, Sindh
Timber	Himalayan foothills, Gujarat

🌐 International trade — the Mesopotamia connection

Mesopotamian texts (which we *can* read) mention trade with three lands. **This is where a readable civilisation tells us about an unreadable one.**

Mesopotamian name	Almost certainly refers to
MELUHHA ★★ ★	The Harappan region (the Indus)
MAGAN	Oman
DILMUN	Bahrain (the intermediary island entrepôt)

Goods Meluhha sent to Mesopotamia: carnelian beads, ivory, copper, gold, shell, wood, and — the poetic one — peacocks and monkeys.

The physical proof:

- **Harappan seals and sealings found in Mesopotamia** (Ur, Kish, Susa)
- **Etched carnelian beads** — a distinctly Harappan technique — found in Mesopotamian royal graves
- **Harappan-style cubical weights** found abroad
- A **Persian Gulf seal** (round, Dilmun type) found at **Lothal** — trade flowing the other way

 **This is exactly the Chapter 1 lesson paying off.** We can't read Harappan script — but Mesopotamians *could* write, and they wrote about trading with Meluhha. **A readable neighbour becomes a window into an unreadable civilisation.** Beautiful methodological point; use it in Mains.

How did goods move? Bullock carts (terracotta cart models are common finds), pack animals, riverboats, and **seagoing ships** — a Mohenjodaro seal actually depicts a boat with a mast.

8 Society and Religion — Tread Carefully Here

This is where UPSC sets its traps, because the evidence is thin and the temptation to over-read is strong.

What the evidence reasonably supports

Observation	Reasonable inference
Different house sizes, some with more rooms and wells	Some social differentiation existed
Citadel vs lower town	Some functional and possibly social separation
Burials with pottery, ornaments, occasionally copper mirrors	Belief in an afterlife ; modest grave goods
Great Bath, and bathrooms in most houses	Ritual purity / cleanliness mattered greatly
Mother Goddess terracotta figurines	Possibly a fertility cult
Pashupati seal	Possibly a proto-Shiva / yogic tradition
Pipal tree motifs on seals; animal motifs	Nature worship , tree and animal veneration
Fire altars at Kalibangan and Lothal	Some form of fire ritual

✗ What we do NOT have — the elimination checklist

Memorise this list. **Prelims questions are frequently built by asserting one of these.**

✗ NOT found	Why it matters
No temples	Unlike Mesopotamia's ziggurats and Egypt's temples. No structure is <i>conclusively</i> a temple
No palaces	No royal residence identified. We do not know how they were governed
No royal names, no king lists	Contrast with Egypt's pharaohs
No monumental tombs	Burials are relatively simple. No pyramids
No large-scale weapons or fortification for war	Walls exist, but few weapons of war (no swords). Little evidence of organised warfare
No horses (reliably), no horse-drawn chariots	Central to Rigvedic culture, absent here
No iron	Bronze Age civilisation
No coins/money	Trade was barter-based, mediated by standard weights
No deciphered script	So no direct knowledge of their beliefs, laws or politics

 **The "who ruled?" question.** NCERT frames it honestly — we genuinely don't know. Three hypotheses circulate:

1. **A single powerful state** (uniformity of bricks, weights, planning suggests strong central control)
2. **Several city-states** sharing a common culture and standards
3. **A collective/oligarchic or priestly authority** with no single monarch

The correct exam answer acknowledges the uncertainty. Overconfidence here is a wrong answer.

 The uniformity is exactly what you'd get from a **shared standards specification** — which could come from a central authority *or* from a widely-adopted convention among peers. **You cannot distinguish the two from the artefacts alone.** That's the honest state of the field.

The End — Why Did It Collapse?

Around **1900 BCE (~3,900 years ago)** the mature Harappan civilisation unravelled. Cities were abandoned, the script vanished, weights and seals fell out of use, long-distance trade dried up, and people dispersed into smaller village settlements.

What we observe archaeologically

- Drains and street cleaning stop being maintained; rubbish accumulates in streets
- New, flimsier structures built over older planned ones — often encroaching on streets
- Craft specialisation declines; the distinctive pottery, seals and weights disappear
- Writing disappears entirely
- **Population shifts eastward and southward** toward the Ganga-Yamuna region and Gujarat

The competing theories

Theory	The argument	The problem with it
 River change / drying ★ (strongest today)	The Ghaggar-Hakra system dried up or shifted course. Over half of all Harappan sites lay along it. The Indus also changed course and flooded	Best supported by geology and site distribution; explains the <i>scale</i> of abandonment
 Climate change ★	Weakening of the monsoon around 2200–1900 BCE (a well-documented global arid event) → agriculture fails → surplus collapses → cities can't be fed	Strong palaeoclimate support; probably works <i>together</i> with the river theory
 Deforestation & environmental degradation	Baking millions of bricks and smelting copper consumed enormous fuel wood → deforestation → soil erosion, flooding, reduced rainfall	Plausible contributory factor
 Overgrazing / soil exhaustion	Land degraded by intensive use	Contributory
 Floods & tectonic shifts	Repeated devastating floods at Mohenjodaro; tectonic uplift damming rivers	Evidence of flooding is real but localised
 Collapse of trade	Mesopotamian demand fell; the trade economy that sustained the cities failed	Contributory, not sufficient alone
 "Aryan invasion" ✗	Proposed by Mortimer Wheeler from some unburied skeletons at Mohenjodaro	Largely REJECTED. The skeletons are from different periods and show no battle wounds; there is no destruction layer, no burnt cities, no mass graves. Do not present this as the accepted view

★ **The mainstream consensus today: a combination** — primarily **climatic drying and hydrological change** (the Ghaggar-Hakra failing), compounded by environmental degradation and collapsing trade. **Gradual decline and de-urbanisation, not a single catastrophic event or invasion.**

 **The exam-safe answer:** "A multi-causal, gradual de-urbanisation, in which climatic aridification and river-course changes were probably decisive; the invasion hypothesis is no longer accepted." That sentence is both true and well-calibrated.

Late Harappan and continuity

The civilisation didn't vanish — it **transformed**. The **Late Harappan** phase (~1900–1300 BCE) shows smaller, rural settlements continuing, with cultural continuity in pottery, some crafts and agricultural practice. Many Harappan traits (weights, bead-making, cotton, urban water management, possibly some religious motifs) resurface in later Indian culture.

 **A comparison worth holding:** Egypt and Mesopotamia recovered and continued as literate urban societies. **India's urbanism disappeared for over a thousand years** — until the "Second Urbanisation" of the Ganga valley around 600 BCE (Chapter 5). That millennium-long gap is a genuinely distinctive feature of Indian history.

Prelims — the highest-yield checklist in the whole book

Topic	The exact thing to know
Site-feature matching ★★ ★	The #1 format. Master the site table
Extremities	Manda (N), Daimabad (S), Alamgirpur (E), Sutkagendor (W)
What they did NOT have ★★ ★	No iron, no horses (reliably), no temples, no palaces, no coins, no royal names
Great Bath	Mohenjodaro, citadel, gypsum mortar + bitumen
Dockyard	Lothal
Ploughed field	Kalibangan (furrows at right angles). Plough model = Banawali
No citadel	Chanhudaro
Three-part city + signboard	Dholavira (also UNESCO 2021)
Fire altars	Kalibangan and Lothal
Meluhha / Magan / Dilmun	Indus / Oman / Bahrain
Weights	Chert, cubical, binary (16 is key)
Faience	An artificial material
Artefacts + material	Dancing Girl (bronze), Priest-King (steatite), Male Torso (red sandstone), Pashupati seal (steatite)
Script	~375–400 signs, right-to-left, boustrophedon , undeciphered
Excavators	Harappa = Daya Ram Sahni (1921) ; Mohenjodaro = R. D. Banerji (1922) ; announced by John Marshall
Decline	Multi-causal; invasion theory rejected

⚠ The three classic traps:

1. "Harappans used horse-drawn chariots in warfare" → **FALSE** (asked in 2013)
2. "Harappans possessed great palaces and temples" → **FALSE** (same question)
3. "Harappans used iron tools" → **FALSE** (Bronze Age)

Mains — how this actually appears

- "Discuss the salient features of Harappan town planning and what they reveal about the nature of political authority."
- "The Harappan civilisation's approach to water management holds lessons for contemporary India. Comment." → **Dholavira's reservoirs, Lothal's dockyard, universal drainage, wells.** A genuinely strong, specific answer.
- **Art & Culture:** seals, the Dancing Girl, terracotta figurines, bead technology, faience.
- **Methodology:** "How does the undeciphered nature of the Harappan script constrain our understanding?" — use the Meluhha/Mesopotamia workaround.

🔗 Current-affairs linkages worth knowing

- **Dholavira — UNESCO World Heritage Site (2021)**, India's 40th
- **Rakhigarhi ancient DNA study (2019)** — found no significant Steppe ancestry in the sampled Harappan individual, feeding the Aryan migration debate. Handle neutrally and factually

- Harappan sites and **ASI conservation** issues; Mohenjodaro's deterioration
- **Water harvesting** policy discussions routinely cite Dholavira

Glossary

Term	Meaning
Citadel	The smaller, higher, walled western part of a Harappan city
Lower Town	The larger, lower, eastern residential and craft area
Great Bath	Large watertight public tank on Mohenjodaro's citadel
Faience	Artificial material of ground sand/quartz, glazed and fired
Carnelian	Reddish semi-precious stone used for beads
Steatite	Soapstone — soft to carve, hardens on firing; used for seals
Bitumen	Natural tar; waterproofing agent in the Great Bath
Gypsum mortar	Binding material used with bricks for waterproofing
Boustrophedon	Writing that alternates direction line by line, "as the ox ploughs"
Meluhha	Mesopotamian name for the Harappan region
Magan / Dilmun	Oman / Bahrain
Sealing	The clay impression left by pressing a seal
Chert	Hard stone used for weights and blades
Chalcolithic	Copper-Stone Age
Proto-history	Period with writing that we cannot read
Lost-wax casting (<i>cire perdue</i>)	Bronze-casting technique used for the Dancing Girl

90-Second Recap

1. **~2600–1900 BCE**, the largest Bronze Age civilisation by area (~1.5 million sq km), contemporary with Egypt and Mesopotamia.
2. **Discovered 1921–22**: Harappa by **Daya Ram Sahni**, Mohenjodaro by **R. D. Banerji**; announced by **John Marshall**.
3. **City plan**: citadel (west, higher, walled) + lower town (east, larger, grid streets).
4. **Drainage is the signature achievement** — covered, sloped, universal, with manholes and soak pits — and **drains were planned before houses**, implying a real civic authority.
5. **Star structures**: Great Bath (Mohenjodaro), dockyard (Lothal), three-part city + signboard + reservoirs (Dholavira), ploughed field + fire altars (Kalibangan), no citadel (Chanhudaro).
6. **Extremities**: Manda (N), Daimabad (S), Alamgirpur (E), Sutkagendor (W); Shortughai in Afghanistan.
7. **Crops**: wheat, barley, pulses, sesame, mustard, **cotton** (world's first); rice at Lothal/Rangpur.
8. **Technology**: bronze, gold, silver, **faience** (man-made), bead-making, shell work, cotton textiles. **NO IRON**.
9. **Seals** of steatite with animal motifs; script of ~375–400 signs, right-to-left, boustrophedon, **undeciphered**.
10. **Weights** of chert, cubical, in a **binary** sequence with 16 as a key unit — standardised across the whole civilisation.

11. **Trade** with **Meluhha/Magan/Dilmun** (Indus/Oman/Bahrain); raw materials sourced from Afghanistan (lapis, tin), Rajasthan (copper), Karnataka (gold), Gujarat (carnelian, shell).
 12. **No temples, palaces, royal names, monumental tombs, or clear evidence of warfare** — governance remains unknown.
 13. **Decline ~1900 BCE:** multi-causal — river drying (Ghaggar-Hakra), climate aridification, environmental degradation, trade collapse. **Invasion theory rejected.**
 14. **Urbanism then disappeared from India for ~1,000 years** until the Ganga valley's Second Urbanisation.
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Test Yourself

Prelims style

1. Which of the following characterises the people of the Indus Civilisation?
 1. They possessed great palaces and temples.
 2. They worshipped both male and female deities.
 3. They employed horse-drawn chariots in warfare. → *Only 2. (This is UPSC 2013, almost verbatim.)*
2. Consider the following statements:
 1. Kalibangan has yielded evidence of a ploughed field.
 2. Banawali has yielded a terracotta model of a plough.
 3. Chanhudaro is the only major Harappan city without a citadel. → *All three are correct.*
3. In Mesopotamian records, "Meluhha" most probably refers to — → *The Harappan/Indus region.* (Magan = Oman, Dilmun = Bahrain.)
4. Consider the following statements about Dholavira:
 1. It is divided into three parts, unlike the usual two.
 2. It is famous for a large inscription of ten Harappan signs.
 3. It was inscribed as a UNESCO World Heritage Site in 2021. → *All three.*
5. Which of the following was NOT known to the Harappans? → *Iron.* (They knew copper, bronze, gold, silver, faience.)

Mains style

6. *"Harappan town planning reveals more about their civic organisation than their political system."* Discuss. **(150 words)**
7. Discuss the water management practices of the Harappan civilisation and their relevance to contemporary India. **(250 words)**
8. Evaluate the various explanations advanced for the decline of the Harappan civilisation. **(250 words)**

Self-check (out loud, no notes):

- What is in the citadel, and what is in the lower town?
 - Why does "drains were built before houses" matter so much?
 - Name five sites and one unique feature of each.
 - Give three reasons the script remains undeciphered.
 - List five things the Harappans did **not** have.
 - Why is the invasion theory rejected?
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